

Unit 8 Practice Problems

Circles Practice

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Unit 8 Practice Problems

Directions

Work all problems first. Then check the answer bank. Detailed solutions are included after the answer bank.

Problems

1. Write the equation of a circle with center $(3, -2)$ and radius 5.
2. Find the center and radius of $(x + 4)^2 + (y - 1)^2 = 49$.
3. A circle has center $(0, 0)$ and radius 10. Is point $(6, 8)$ inside, outside, or on the circle?
4. A circle has center $(0, 0)$ and radius 6. Is point $(3, 4)$ inside, outside, or on the circle?
5. Find the radius and equation of a circle with center $(2, 1)$ passing through $(8, 9)$.
6. Find the equation of the tangent line to the circle centered at $(0, 0)$ at point $(3, 4)$.
7. Find the equation of the tangent line to the circle centered at $(2, -1)$ at point $(6, 2)$.
8. A central angle intercepts an arc of 110° . Find the central angle.
9. An inscribed angle intercepts an arc of 124° . Find the angle.
10. An inscribed angle measures 38° . Find its intercepted arc.
11. A tangent-chord angle intercepts an arc of 92° . Find the angle.
12. An exterior angle formed by two secants intercepts arcs of 150° and 70° . Find the angle.
13. Two chords intersect inside a circle. The intercepted arcs are 80° and 120° . Find the angle.
14. A cyclic quadrilateral has one angle 105° . Find its opposite angle.
15. Intersecting chords have segment products $6 \cdot 8 = x \cdot 12$. Find x .
16. Two secants are drawn from an external point. One has external part 4 and internal part 11. The other has external part 5 and internal part x . Find x .
17. A tangent and secant are drawn from an external point. The secant has external part 6 and internal part 18. Find the tangent length.
18. What angle does a radius form with a tangent at the point of tangency?
19. A chord has length 16, and the distance from the center to the chord is 6. Find the radius.
20. Two chords in the same circle are both 5 cm from the center. What can you conclude?
21. Explain why all circles are similar.
22. What is the measure of an angle inscribed in a semicircle?
23. Which construction locates a triangle's circumcenter?
24. Which construction locates a triangle's incenter?
25. A regular hexagon is inscribed in a circle of radius 10. Find the side length of the hexagon.

Answer Bank

1. $(x - 3)^2 + (y + 2)^2 = 25$.
2. Center $(-4, 1)$, radius 7.
3. On the circle.
4. Inside the circle.
5. Radius 10; $(x - 2)^2 + (y - 1)^2 = 100$.
6. $y - 4 = -\frac{3}{4}(x - 3)$.
7. $y - 2 = -\frac{4}{3}(x - 6)$.
8. 110° .
9. 62° .
10. 76° .
11. 46° .
12. 40° .
13. 100° .
14. 75° .
15. 4.
16. 7.
17. 12.
18. 90° .
19. 10.
20. They are congruent.
21. A translation aligns centers and a dilation scales one radius to the other.
22. 90° .
23. Perpendicular bisectors of the sides.
24. Angle bisectors of the angles.
25. 10.

Detailed Solutions

1. Use $(x - h)^2 + (y - k)^2 = r^2$. With center $(3, -2)$ and radius 5, the equation is $(x - 3)^2 + (y + 2)^2 = 25$.
2. Compare to $(x - h)^2 + (y - k)^2 = r^2$. Since $(x + 4) = (x - (-4))$, center is $(-4, 1)$, and $r = \sqrt{49} = 7$.
3. Compute distance from center: $\sqrt{6^2 + 8^2} = 10$. This equals the radius, so the point is on the circle.
4. Distance from center is $\sqrt{3^2 + 4^2} = 5$, which is less than 6. The point is inside the circle.
5. Radius is the distance from $(2, 1)$ to $(8, 9)$: $\sqrt{6^2 + 8^2} = 10$. Equation: $(x - 2)^2 + (y - 1)^2 = 100$.
6. Radius slope from $(0, 0)$ to $(3, 4)$ is $4/3$. The tangent slope is $-3/4$. Through $(3, 4)$, the tangent is $y - 4 = -\frac{3}{4}(x - 3)$.
7. Radius slope from $(2, -1)$ to $(6, 2)$ is $\frac{3}{4}$. The tangent slope is $-\frac{4}{3}$. Through $(6, 2)$, the tangent is $y - 2 = -\frac{4}{3}(x - 6)$.
8. A central angle has the same measure as its intercepted arc, so the angle is 110° .
9. An inscribed angle is half its intercepted arc: $124/2 = 62^\circ$.
10. An inscribed angle is half its arc, so the arc is $2(38) = 76^\circ$.
11. A tangent-chord angle is half its intercepted arc: $92/2 = 46^\circ$.
12. An exterior secant angle is half the difference of arcs: $\frac{1}{2}(150 - 70) = 40^\circ$.
13. An angle formed by intersecting chords inside a circle is half the sum of intercepted arcs: $\frac{1}{2}(80 + 120) = 100^\circ$.
14. Opposite angles of a cyclic quadrilateral are supplementary. The opposite angle is $180 - 105 = 75^\circ$.
15. Use intersecting chords: $6 \cdot 8 = x \cdot 12$. Then $48 = 12x$, so $x = 4$.
16. Use secant-secant: external times whole. First whole length is $4 + 11 = 15$, so product is $4(15) = 60$. Second whole is $5 + x$, so $5(5 + x) = 60$. Then $25 + 5x = 60$, giving $x = 7$.
17. Use tangent-secant: $t^2 = (\text{external})(\text{whole})$. Whole secant is $6 + 18 = 24$. Thus $t^2 = 6(24) = 144$, so $t = 12$.
18. A radius to a tangent point is perpendicular to the tangent, so the angle is 90° .
19. The perpendicular from the center bisects the chord, so half the chord is 8. Use a right triangle: $r^2 = 6^2 + 8^2 = 100$, so $r = 10$.
20. Chords equidistant from the center of the same circle are congruent.
21. Move one circle's center to the other's center by translation, then dilate by the ratio of the radii. Since this maps one circle onto the other, all circles are similar.
22. An angle inscribed in a semicircle intercepts a diameter, so it is a right angle: 90° .
23. The circumcenter is located by constructing perpendicular bisectors of the sides of the triangle.
24. The incenter is located by constructing the angle bisectors of the triangle.
25. A regular hexagon inscribed in a circle is made of six equilateral triangles with side length equal to the radius, so the side length is 10.